

### EXECUTIVE SUMMARY

**Industry** - Healthcare

**Segment** - Image Processing

**Services** - Image Processing  
and Prediction Model

**Technologies**  
Python, YoloV5

*"Our POC project will help  
reduce the error of  
misdiagnosis"*

*The use of artificial intelligence (AI) in diagnostic medical imaging is undergoing extensive evaluation. AI has shown impressive accuracy and sensitivity in the identification of imaging abnormalities.*

*Among the most promising clinical applications of AI is diagnostic imaging, and mounting attention is being directed at establishing and fine-tuning its performance to facilitate detection and quantification of a wide array of clinical conditions. Investigations leveraging computer-aided diagnostics have shown excellent accuracy, sensitivity, and specificity for the detection of small radiographic abnormalities, with the potential to improve public health.*

*Our solution is a data-driven approach for automatic prediction of deterioration risk using a deep neural network that learns from chest X-ray images.*

### PROBLEM SUMMARY

Doctors diagnose and treat medical conditions using imaging techniques like CT and PET scans, MRIs, and X-rays. It might happen when working with such a wide variety of medical tools, radiologists face many daily challenges, perhaps the most difficult being the chest radiograph.

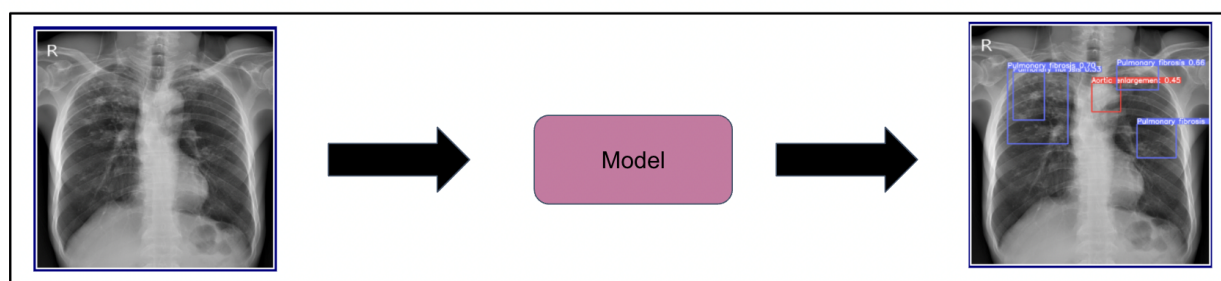
The interpretation of chest X-rays can lead to a medical misdiagnosis, even for the best practising doctor.

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***This use case is an object detection task to find the class and location of thoracic abnormalities from chest x-ray images.***

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### SOLUTION SUMMARY



- Image tells about, how the algorithm detect the class and location of thoracic abnormalities from chest x-rays
- The dataset used comprises 18,000 poster anterior (PA) CXR scans in DICOM format, where we consider 14 critical radiographic findings

- The AI analyzes unstructured medical data (radiology images, blood tests, EKGs, Genomics, patient medical history) to give doctors better insight into a patient's real-time needs
- This above AI system will be a one stop solution challenges mentioned in problem summary.

### **BENEFITS**

- Earlier disease detection with AI
- Availability of rural residents to access quality healthcare
- Prevent Death and critical medical conditions due to misdiagnosing illness and medical error
- Minimise Deadly human errors in diagnosing
- Avoid Lengthy hospital procedures
- Early detection of diseases
- One click identification of abnormalities expected in chest x-rays, like Aortic enlargement, Atelectasis, Calcification, Consolidation, Interstitial lung disease (ILD), Infiltration etc.